

Nagarjun Mallesh

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 Expected CTC: ₹30-40 LPA |  Availability: Immediate Joiner |  Status: Based in Bangalore (No relocation needed)

PROFESSIONAL SUMMARY

Backend Software Engineer with 4+ years of experience building **highly reliable, scalable enterprise applications** serving millions of users. Expertise in **Java, Spring Boot, microservices architecture**, and distributed systems deployed at massive scale (100,000+ endpoints). Recently returned to Bangalore after gaining experience with US-based product companies, bringing a proven track record of delivering **99.9% uptime** systems and leading cross-functional teams. Strong foundation in cloud-native architectures (AWS), event-driven systems, and production deployment best practices.

Core Expertise: Java & Spring Boot | Microservices Architecture | Distributed Systems | Enterprise Scale | RESTful APIs | AWS Infrastructure | Cross-functional Leadership | Code Review & Mentoring

EXPERIENCE

Senior Systems Analyst | ML Technologies LLC, Boston, MA | Jul 2025 - Nov 2025 (Relocated to India for family reasons)
US-based ML consultancy specializing in emergency services optimization

- Architected and deployed **GenAI-powered vehicle recovery system** using AWS Bedrock (Claude) and OpenSearch vector database, reducing emergency response times by 70% and processing 5,000+ daily requests across 12 metropolitan areas
- Established a technical roadmap for GenAI adoption and **mentored three junior engineers** on RAG best practices, LangChain development, and prompt engineering, accelerating team delivery velocity by 40%
- Designed a **serverless emergency services platform** with API Gateway + Lambda architecture, processing 150,000+ monthly API requests with 99.9% uptime and auto-scaling capabilities
- **Tech Stack:** Python, AWS Bedrock, OpenSearch, Lambda, API Gateway, LangChain, CloudFormation

Backend Software Engineer | Keelworks Foundation, Oak Harbor, WA | Jul 2024 - Jul 2025
US-based non-profit foundation focused on volunteer management technology.

- **Architected and developed Spring Boot backend services** for a volunteer management platform serving 10,000+ active users with 92% satisfaction score and 99.9% uptime
- **Led cross-functional team** of 4 engineers and QA specialists through complete SDLC, designed RESTful APIs for mobile and web clients, implemented JWT-based authentication and role-based access control
- **Designed event-driven architecture** using AWS EventBridge and Lambda for workflow automation, achieving 60% cost reduction compared to an EC2-based solution while improving scalability and reliability
- **Optimized database performance** working with PostgreSQL on AWS RDS Multi-AZ, designed efficient schemas, implemented connection pooling, and query optimization, reducing response times by 40%
- **Tech Stack:** Java, Spring Boot, PostgreSQL, AWS (Lambda, EventBridge, RDS, S3, CloudWatch), Terraform, Docker, REST APIs

Software Engineer | Hewlett-Packard Inc, Bangalore, India | Oct 2019 - Jul 2022
Global technology company - BIOS/Firmware automation and management platform development

- Delivered **global automation solution** by developing PowerShell-based BIOS configuration system deployed across HP manufacturing units in 8 countries, transforming 4-day manual processes into 15-minute automated workflows and eliminating 90% of configuration errors
- **Led Infrastructure Automation Project** for AWS-based OS deployment system, reducing provisioning time from 3 days to 4 hours, successfully deployed to 100,000+ client machines globally
- Built a **Management Dashboard Platform** by orchestrating a cross-functional team of 6 engineers and QA to architect a React.js + Node.js web application for real-time BIOS monitoring across 50,000+ devices
- **Tech Stack:** PowerShell, C#, Node.js, React.js, AWS (EC2, S3, Lambda), Python, REST API

Associate Software Engineer | Digital API Craft Pvt Ltd, Bangalore, India | Jul 2019 - Oct 2019
Digital transformation consultancy - Banking sector projects

- Executed **microservices migration** for banking application using Spring Boot and Apache Kafka, handling 50,000+ daily transactions with 99.8% reliability and reducing response latency by 35%
- Built a **domain provisioning automation system** using Golang with Route53 DNS integration, automating the client onboarding process and reducing manual errors by 90%
- **Tech Stack:** Java, Spring Boot, Golang, Apache Kafka, PostgreSQL, AWS Route53

SKILLS

Programming Languages: Java, JavaScript/TypeScript, Node.js, Python, Golang, C#, PowerShell

Java & Backend Frameworks: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, Node.js/Express, FastAPI, Flask

APIs & Integration: RESTful APIs, GraphQL, Microservices Architecture, OpenAPI/Swagger, JWT Authentication, API Gateway

Cloud & Infrastructure: AWS (EC2, RDS, Lambda, API Gateway, EventBridge, S3, CloudWatch, Step Functions, VPC, ALB) | Docker | Kubernetes | Terraform | CI/CD (GitHub Actions, Jenkins)

Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Apache Kafka

Architecture & Design: Microservices Architecture | Event-Driven Systems | Distributed Systems | Multi-tenant SaaS | High Availability Design | System Design

Development Practices: Agile/Scrum | Code Review | Test-Driven Development | CI/CD Pipelines | Blue-Green Deployments | Production Monitoring

EDUCATION

Master of Science in Information Systems | Northeastern University, Boston, USA

Sep 2022 - May 2024 | GPA: 3.6/4.0

Bachelor of Engineering in Information Science | Visvesvaraya Technological University, India

Aug 2015 - Jun 2019 | GPA: 8.0/10.0

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **Introduction to Artificial Intelligence (CS50 AI) | Harvard University - 2025**
- **AWS Essential Training for Developers | LinkedIn Learning - 2021**
- **Learning Data Science: Understanding the Basics | LinkedIn Learning - 2021**
- **Learning Git and GitHub | LinkedIn Learning - 2021**
- **Introduction to Big Data | Coursera - 2020 (Completed during HP tenure)**

Academic Coursework: Machine Learning, Deep Learning, Cloud Computing, Distributed Systems (Northeastern University, 2022-2024)

NOTABLE PROJECTS

AI Travel Matching Platform | Academic Capstone Project

Developed a **Spring Boot web application** with Binary Search Tree algorithms for optimized travel companion matching, integrated with OpenAI GPT-4 for intelligent profile analysis. Served 500+ users during pilot phase with 85% successful match rate.

- **Tech Stack:** Java, Spring Boot, OpenAI API, PostgreSQL, React.js
- **Features:** Real-time matching algorithm, AI-powered compatibility scoring, Secure authentication

Multi-Tenant Student Housing Platform | Academic/Personal Project

Architected a **full-stack Spring Boot application** connecting international students with landlords through real estate consultants. Built a multi-tenant SaaS platform with role-based dashboards for students, property owners, and agents. Implemented **Spring Security** for authentication, **Spring Data JPA** with **PostgreSQL** for data persistence, and **Thymeleaf** with **jQuery** for a responsive UI.

- **Tech Stack:** Java, Spring Boot, Thymeleaf, jQuery, PostgreSQL, AWS (EC2, ALB, API Gateway, Step Functions, VPC)
- **Impact:** Served 500+ users across multiple tenants, automated emergency contact workflows, and reduced property matching time by 60%

Concurrent Text Normalization Engine | Freelance Project

Built a multi-threaded document processing system leveraging AWS Bedrock LLMs to normalize 10,000+ legal documents simultaneously with 95% accuracy. Implemented producer-consumer pattern with thread pools for optimal resource utilization, reducing processing time from 8 hours to 45 minutes.

- **Tech Stack:** Python, Threading, AWS Bedrock, S3, Lambda, CloudWatch
- **Impact:** 90% reduction in processing time, handling 500GB+ document corpus

TECHNICAL CONTRIBUTIONS & COMMUNITY

Technical Writing: Publish technical articles on Medium covering AI/ML implementation patterns and cloud architecture

Profile: medium.com/@nagarjunmalleesh

Code Quality Leadership: Conducted 50+ code reviews across cross-functional teams, establishing best practices for GenAI implementation and serverless architecture